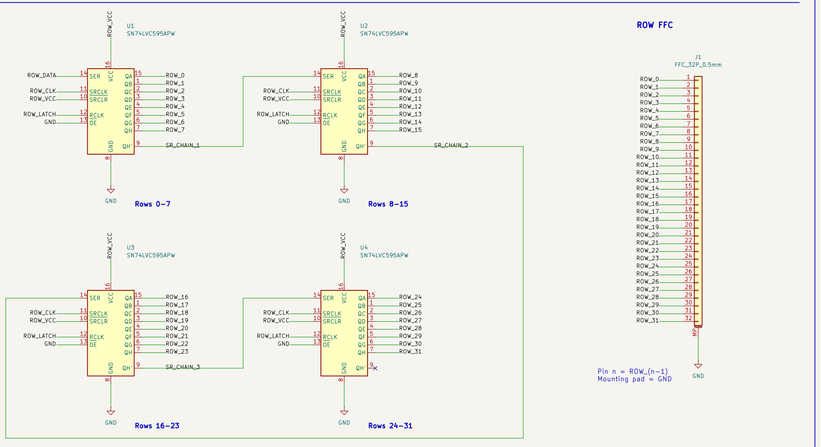
TAXELSCAN / REV 3

32 x 32 matrix > two sense banks > 16-bit ADC > RP2354A > RS-485 harness

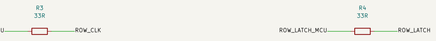
01 ROW EXCITATION

MCU serial stream > four latched drivers > 32-way row FFC



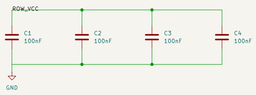
ROW FFC

Serial order: U1 > U2 > U3 > U4
Outputs enabled; unselected rows held LOW.



Clock / latch damping

U1–U4 bypass

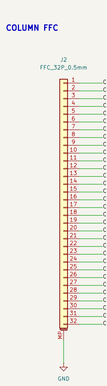


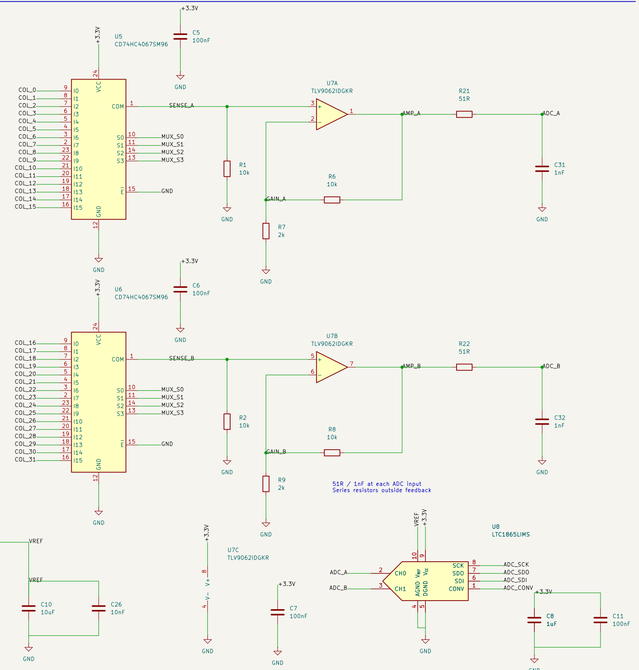
Excitation rail link

02 COLUMN SENSING + CONVERSION

32 columns > two 16:1 muxes > gain of 6 > filtered ADC inputs

COLUMN FFC





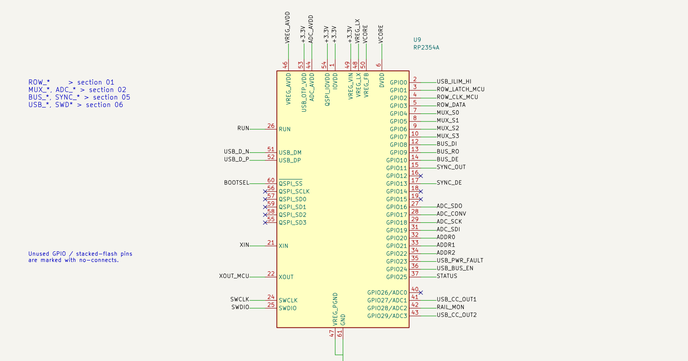
R1/R2: 10k pulldown
Rf/Rg: 10k / 2k (G = 6)

VREF tracks ROW_VCC

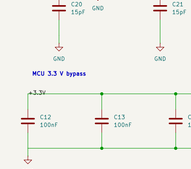
51R / 1nF at each ADC input
Series resistors outside feedback

03 SCAN CONTROLLER

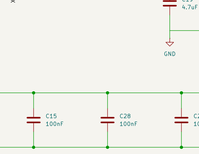
RP2354A / clock / core supply / local bypass



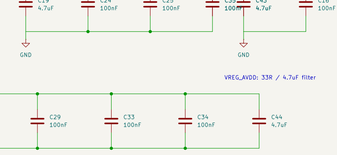
12 MHz / COG load capacitors



R36: 1k XOUT series, limits crystal drive

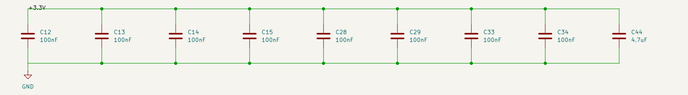


Core regulator / DVDD bypass



MCU 3.3 V bypass

VREG_AVDD: 33R / 4.7uF filter



04 POWER + RAIL MONITOR

Harness or USB power > diode OR > local 3.3 V buck

3.3 V buck output
Feedback: 180k / 40.2k

5 V rail monitor
100k / 47k divider to MCU

MCU analog supply filter

ROW_VCC link / VREF filter:
see sections 01 and 02.
All returns use GND.

05 COMMUNICATIONS + BOARD ADDRESS

Half-duplex data pair and dedicated receive-only sync pair

DATA / half duplex

HARNESS IN

HARNESS OUT

1 +5V 2 GND 3/4 DATA P/N 5/6 SYNC P/N

R11 / R12: fit 120R only
at the two harness ends.

SYNC / master drives DE (GPIO13)
R37 keeps other boards receive-only

Board address straps: ADDR0 / ADDR1 / ADDR2
Closing a strap connects that input to GND.

Status Indicator

06 USB + PROGRAMMING

USB-C data / configuration straps / SWD / BOOTSEL

CC1 / CC2 terminate inside U13 (section 07).
Do not fit external 5.1k pulldowns.

SWD DEBUG

RUN pull-up

D5: USB ESD, D+/D−/VBUS at the connector

BOOTSEL: series resistor limits the flash-CS stub

USB series resistors at MCU end
USB connector shell and all returns: GND

07 USB-C POWER TO THE HARNESS

Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS

UFP / sink only; ADDR left open selects GPIO mode.
LAI variant required: OUT1/OUT2 track current changes.

LOW limit: R27 only, about 0.32 A nominal
HIGH limit: R27 || R28, about 0.63 A nominal
D4 blocks an externally powered harness from USB.

POWER POLICY (GPIO 12 / 13 / 14 / 15 / 23)
Reset / no MCU / Hi-Z = OFF
Default USB current: enable LOW only after
USB configuration grants 500 mA.
CC advertises 1.5 A or 3 A: HIGH permitted.
Fewer boards draw less; limits do not force current.

Firmware must service CC changes promptly,
disable on default-current suspend/reset,
and latch faults OFF until an explicit retry.
See USB_POWER.md for budgets and bring-up.

POWER-NET DRIVE DECLARATIONS

Sensor path: ROW_n > mat element > COL_n > SENSE_A/B > AMP_A/B > ADC_A/B

R1/R2 and gain-network values remain fit-on-test values from the existing design.

Reading the sheet

Matching net labels connect between sections. A dot marks a branch; an X marks an unused pin.
Bypass capacitors are grouped by supply. USB attach current and the complete eight-board
startup / operating budget require measurements; schematic checks cannot establish them.

3.3 V logic | USB-C data and protected chain power | SWD debug
Row drive / dual-bank sensing / RP2354A / RS-485

Sheet: /
File: rev3.kicad_sch

Title: TaxelScan | 32 x 32 sensor interface

Size: User	Date:	Rev: 3
KiCad E.D.A. 10.0.5		Id: 1/1